

Before delving into the world of zero-knowledge, we need to settle the notation which we are going to use throughout the book.

Set Theory

A **set** is a well-defined collection of objects. The objects that belong to a set are called its **elements**. We will denote sets by capital letters, if a is an element of the set A , we write $a \in A$.

Let us enumerate some fundamental sets:

- $\mathbb{N} = \{1, 2, 3, \dots\}$ – a set of natural numbers.
- $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$ – a set of integers.
- $\mathbb{Q} = \{\frac{n}{m} : n \in \mathbb{Z}, m \in \mathbb{N}\}$ – a set of rational numbers.
- $\mathbb{R}$ – a set of real numbers. Examples: 2.2, -3.9 , 0.12, 1.4, -6.7 , $\dots$
- $\mathbb{R}_{>0}$ – a set of positive real numbers. Examples: 2.6, 10.4, 100.2.
- $\mathbb{C}$ – a set of complex numbers¹. Examples: $1 + 2i$, $5i$, $-7 - 5.7i$, $\dots$

To represent the number of elements in a set A , we write $|A|$. If the set is finite, $|A| \in \mathbb{N}$, otherwise $|A| = \infty$. $A \subset B$ denotes “ A is a subset of B ” (meaning that all elements of A are also in B , e.g., $\mathbb{Q} \subset \mathbb{R}$).

$A \cap B$ means the intersection of A and B (a set of elements belonging to both A and B), while $A \cup B$ – the union of A and B (the set of elements belonging to either A or B). $A \setminus B$ denotes the set difference (the set of elements belonging to A , but not B). $\bar{A}$ denotes the complement of A (the set of elements not belonging to A). All operations are illustrated in **Figure 0.1** (this picture is typically called the *Venn Diagram*).

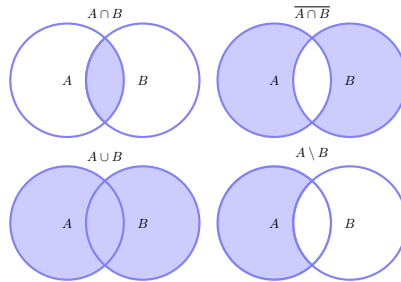


Figure 0.1: Set operations illustrated with Venn diagrams.

As you may have noticed, to define the set we typically write $\{f(a) : \phi(a)\}$, where $f(a)$ is some function and $\phi(a)$ is a predicate (function, inputting a and returning true/false if a certain condition on a is met). For example, $\{x^3 : x \in$

¹Complex number is an expression in a form $x + iy$ for $i^2 = -1$

$\mathbb{R}, x^2 = 4\}$ is “a set of values x^3 which are the real solutions to equation $x^2 = 4$ ”. It is quite easy to see that this set is simply $\{2^3, (-2)^3\} = \{8, -8\}$.

Cartesian Products and Functions

The notation $A \times B$ means a set of pairs (a, b) where $a \in A$ and $b \in B$ (or, written shortly, $A \times B = \{(a, b) : a \in A, b \in B\}$), called a Cartesian product. We additionally introduce notation $A^n := A \times A \times \cdots \times A$ – Cartesian product n times. For example, $\mathbb{Q}^3$ is a set of triplets (a, b, c) where $a, b, c \in \mathbb{Q}$, while $\mathbb{Q}^2 \times \mathbb{R}$ is a set of triplets (a, b, c) where $a, b \in \mathbb{Q}$ and $c \in \mathbb{R}$.

Subsets of $A \times B$ are called relations. We define a function $f \subset A \times B$ from a set A to a set B , as the special type of relation in which for each element $a \in A$ there is a *unique* element $b \in B$ such that $(a, b) \in f$. In other words, for every element in A we assigns a unique element in B . We usually write $f : A \rightarrow B$ to denote such a relation and write $f(a) = b$ instead of writing down $(a, b) \in f$.

We say the function $f : A \rightarrow B$ is **injective** if condition $a_1 = a_2$ implies $f(a_1) = f(a_2)$. Function f is said to be **surjective** if for each $b \in B$ there exists $a \in A$ such that $f(a) = b$. A function that is both injective and surjective is called a **bijection**.

Logic

Statement beginning with $\forall$ means “for all...”. For instance, $(\forall a \in A \subset \mathbb{R}) : \{a < 1\}$ is read as: “For any a in set A (which is a subset of real numbers), it is true that $a < 1$ ”. Or, more shortly, “Any (real) a from A is less than 1”.

Statement beginning from $\exists$ means “there exists such...”. Let us consider the following example: $(\exists \varepsilon > 0)(\forall a \in A) : \{a > \varepsilon\}$ is read as “there exists such a positive ε such that for any element a from A , a is greater than ε ”, or, more concisely, “there exists a positive constant ε such that any element from A is greater than ε ”.

Statement beginning from $\exists!$ means “there exists a unique...”. For example, $(\exists! x \in \mathbb{R}_{>0}) : \{x^2 = 4\}$ is read as “there exists a unique positive real x such that $x^2 = 4$ ”.

Symbol $\wedge$ means “and”. For example, $\{x \in \mathbb{R} : x^2 = 4 \wedge x > 0\}$ is read as “a set of real x such that $x^2 = 4$ and x is positive”. Of course, $\{x \in \mathbb{R} : x^2 = 4 \wedge x > 0\} = \{2\}$.

Symbol $\vee$ means “or”. For example, $\{x \in \mathbb{R} : x^2 = 4 \vee x^2 = 9\}$ is read as “a set of real x such that either $x^2 = 4$ or $x^2 = 9$ ”, this set is equal to $\{\pm 2, \pm 3\}$.

Sequences, Matrices, and Vectors

A **sequence** is a mapping from natural numbers $\mathbb{N}$ to some specific set X . To denote the sequence, we write $\{x_n\}_{n \in \mathbb{N}} \subset X$. If the sequence iterates through some specific set $\mathcal{I} \subseteq \mathbb{N}$, we write $\{x_n\}_{n \in \mathcal{I}}$. For example, $\{x_n\}_{n \in \{1,2,5\}} = (x_1, x_2, x_5)$. We write vectors in bold font, e.g., $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{F}^3$, while

matrices are denoted by capital letters in regular font. Here is an example matrix:

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in \mathbb{F}^{2 \times 2}$. For additional linear algebra notation, refer to ??, where we introduce the very basics of linear algebra.

Finally, we denote the set $\{0, \dots, n-1\}$ for $n \in \mathbb{N}$ by $[n]$. This notation is used very extensively in the book, so make sure to remember it. For example, instead of writing $\sum_{k=0}^{n-1} k$ we sometimes write $\sum_{k \in [n]} k = \frac{n(n-1)}{2}$. Or, instead of specifying n conditions $z_1 = f(x_0), z_2 = f(x_1), \dots, z_n = f(x_{n-1})$ explicitly, we simply say $z_{i+1} = f(x_i)$ for all $i \in [n]$.

Algorithms and Probabilities

Suppose we specified the algorithm A that takes some input x and returns some output y . We write $y \leftarrow A(x)$ to denote that the algorithm A is executed on input x and returns output y . To denote some party in the protocol we use calligraphic font, e.g., $\mathcal{P}$ typically denotes the prover while $\mathcal{V}$ denotes the verifier. Such parties are algorithms themselves, so we can easily write $y \leftarrow \mathcal{P}(x)$ to denote that the prover, based on x , has outputted y .

To denote the probability of an event E happening, we write $\Pr[E]$. For example, if E is an event of rolling a dice and getting a number 4, then $\Pr[E] = \frac{1}{6}$. To denote the random sampling from the finite set X , we write $x \leftarrow \$ X$. For example, $x \leftarrow \$ \{1, 2, 3, 4, 5, 6\}$ denotes the random sampling of x from the set $\{1, 2, 3, 4, 5, 6\}$.

To denote the probability of both E and F happening, we write $\Pr[E, F]$ (as opposed to $E \cap F$). To denote the conditional probability, we write $\Pr[E|F]$, which means the probability of E happening given that F has already happened. In practice, we place the protocol execution in F , while E is the event of the protocol outputting a certain result. For example,

$$\Pr[x \equiv 1 \pmod{3} \mid x \leftarrow \text{RollDice}(\cdot)] = \frac{1}{3}.$$

When event F can be written concisely, we write it in the subindex of $\Pr$. For instance, $\Pr_{x \leftarrow \$[6]}[x \equiv 0 \pmod{2}] = \frac{1}{2}$.